
Grounded Answer Updating for Multi-Timestamp Video Question Answering

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 **Code**

Abstract

Video-language models are often evaluated on questions with a single fixed answer, but many real video tasks are dynamic: the correct answer can change as the scene evolves. A useful model must therefore answer from the current state of the video rather than from an earlier observation. In this work, we study this problem in multi-timestamp video question answering, where related questions are asked at different moments in the same video and later questions are harder than earlier ones.

We propose a structured answer-updating framework for dynamic video understanding. Given frames sampled from a video prefix, the model is prompted to produce a single JSON output that answers a three-question group and provides a short rationale for each answer. We then train the model with Group Relative Policy Optimization (GRPO) using a reward that first enforces valid structured output and then gradually shifts from partial question-level correctness to full group-level correctness. This design encourages the model to produce answers that are both well-formed and consistent across the question hierarchy.

We instantiate the framework with Qwen2.5-VL-7B and evaluate three settings: a standard free-form baseline, structured JSON prompting without training, and GRPO training with the grouped JSON format. Structured prompting alone yields a strong improvement over the free-form baseline. GRPO training brings further gains and gives the best performance after five epochs, improving overall accuracy from 38.57% to 56.37% and score from 6.81 to 19.90. The gains are especially large on reasoning-heavy and event-centric cases. We also find that after training, the model answers the hardest question more reliably when earlier questions in the same group are correct, suggesting stronger hierarchical consistency. These results show that structured answer updating and reward-based optimization are effective tools for improving temporal reasoning in dynamic video understanding.

1 Introduction

Video-language models have made strong progress on perception, understanding, and reasoning over long visual contexts. Yet many evaluation settings still assume that each video question has a single fixed answer. This assumption is too limited for dynamic environments. In driving, sports, egocentric video, and interactive settings, the scene changes continuously, and the correct answer can change with it. A model that answers from an earlier observation rather than the current state may perform well on standard video QA while still failing in temporally grounded decision making.

Recent work has begun to highlight this gap more clearly. StreamingBench shows that current multimodal large language models still perform far below humans when questions are asked

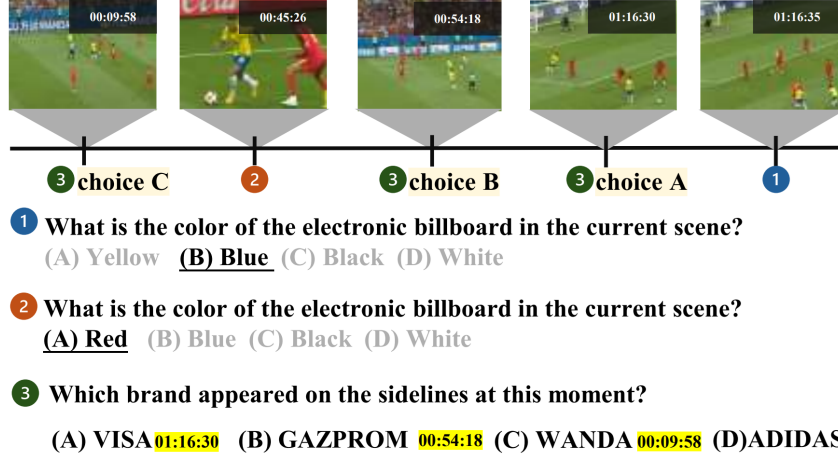


Figure 1: Example of multi-timestamp QA in RTV-Bench. Although the semantic question remains the same across queries, the correct answer is defined by the scene state at the queried timestamp. The figure highlights that models must ground each prediction in the current visual evidence and update the answer when the scene changes, instead of relying on stale observations from earlier timestamps.

during a stream rather than after the full video is available, especially when visual, audio, and contextual signals must be handled together Lin et al. [2024]. OVBench and VideoChat-Online further emphasize that online video understanding requires reasoning over past, current, and future temporal contexts, together with memory for earlier events and anticipation of what comes next Huang et al. [2025]. At the system level, InternLM-XComposer2.5-OmniLive separates perception, memory, and reasoning for long-term streaming interaction, arguing that continuous multimodal understanding is difficult to achieve with a single monolithic context-only architecture Zhang et al. [2024]. VITA-1.5 studies real-time vision and speech interaction and shows that adding new interactive capabilities requires careful training design in order to preserve existing multimodal performance Fu et al. [2025]. Together, these works mark a broader shift from offline video comprehension toward models that must operate under evolving temporal context.

RTV-Bench was introduced to study this challenge directly in multi-timestamp video question answering Xun et al. [2026]. Its core design revisits related questions at different moments in the same video and evaluates whether a model can maintain an updated representation of the scene as visual evidence evolves. This benchmark exposes a central problem for video-language models: *answer updating*. The model must identify what in the scene is relevant, determine whether that state has changed, and produce an answer grounded in what is true at the queried moment. Failures are not limited to low-level perception. A model may retain a stale answer after the scene changes, update an answer when no update is needed, or produce a plausible explanation that is not supported by the current video evidence.

A natural way to address this problem is to make temporal grounding explicit in the model output. Rather than asking the model to emit only a final option letter, we require a structured response that separates the answers to related questions and provides a short grounded justification for each one. This has two advantages. First, it makes the model’s behavior easier to inspect. Second, it creates an intermediate target that can be optimized during training. In language reasoning, reinforcement learning has shown that structured intermediate behavior can improve when the reward is aligned with the desired reasoning outcome Shao et al. [2024]. This suggests a similar direction for dynamic video understanding: reward the model not only for the final answer, but also for producing a valid and consistent structured response across temporally related questions.

In this work, we study structured answer updating for multi-timestamp video question answering. We focus on grouped question sets in which two earlier questions are followed by

a more difficult third question. Our method prompts the model with sampled frames from a video prefix and requires a single JSON output containing answers for the full question group, together with a short rationale for each answer. We then train the model with Group Relative Policy Optimization (GRPO) under a strict schema-validity constraint and a curriculum reward. Early in training, the reward gives partial credit for correct answers to individual questions. Later, it places increasing weight on solving the full group correctly. This design encourages the model to first learn stable structured output and then learn hierarchical consistency across the question group.

We instantiate the framework with Qwen2.5-VL-7B and evaluate three settings: a standard free-form baseline, structured JSON prompting without additional training, and GRPO training with the grouped JSON schema. The results show a clear progression. Structured prompting alone improves performance over the free-form baseline, indicating that explicit output structure already helps temporal grounding. GRPO training produces further gains and yields the best results after five epochs. The strongest improvements appear in reasoning-heavy and event-centric cases, where correct answer updating is most critical. We also find that after training, the model answers the hardest question more reliably when earlier questions in the same group are correct, suggesting improved hierarchical consistency.

Our contributions are threefold. First, we formulate answer updating as a structured output problem for dynamic video question answering and design a grouped JSON representation that makes temporal grounding explicit. Second, we introduce a GRPO-based training method with a schema-gated curriculum reward that shifts from partial correctness to full group consistency. Third, we provide empirical evidence on RTV-Bench that structured prompting and reward-based optimization together produce substantial gains over a strong free-form baseline. These results support a simple claim: in dynamic video understanding, better temporal reasoning requires not only better answers, but better mechanisms for updating them.

2 Method

2.1 Problem Setting

We study answer updating in dynamic video question answering. The input is a video prefix from time 0 up to a queried moment, represented by sampled frames, together with a grouped multiple-choice question set. Each group contains three related questions of increasing difficulty. The model must answer each question using the scene state at that question’s own queried timestamp rather than relying on stale observations from earlier moments.

This setting is difficult for two reasons. First, the relevant visual state may change over time, so the correct answer is not fixed across the video. Second, later questions in a group are harder and often depend on the understanding required for earlier questions. A model must therefore maintain temporal grounding and hierarchical consistency at the same time.

2.2 RTV-Bench

We evaluate our method on RTV-Bench, a benchmark designed to measure continuous perception, understanding, and reasoning in real-time video settings Xun et al. [2026]. RTV-Bench was introduced to address a gap in existing video QA benchmarks, which usually assume a single static answer for each question and therefore do not directly test whether a model can update its beliefs as the scene evolves. RTV-Bench instead revisits the same conceptual query at different moments in the same video, so the correct answer may change with time Xun et al. [2026].

The benchmark is built around three design principles. First, it uses *multi-timestamp question answering*, where the same underlying query is evaluated at multiple points in a video. Second, it uses a *hierarchical question structure*: each question set contains about three multiple-choice questions, with the first two being simpler and the final one being more complex and requiring broader context or more integrated reasoning. Third, it reports *multi-dimensional evaluation* across eight capability dimensions: Temporal Perception, Scene

Perception, Visual Perception, Future Prediction, Phenomenological Understanding, Intent Analysis, Global Understanding, and Spatiotemporal Reasoning Xun et al. [2026].

RTV-Bench contains 552 videos collected from dynamic domains such as driving, sports, and egocentric video, with a total duration of 167.2 hours and 4.6K manually curated QA pairs Xun et al. [2026]. The dataset was constructed through manual video collection and filtering, LLM-assisted template generation, and human refinement of each question and timestamp. Annotators explicitly crafted questions whose correct answers evolve over time and determined the earliest valid timestamp associated with each answer option. This construction process is important for our work because it turns answer updating into a first-class evaluation target rather than a side effect of long-video understanding.

RTV-Bench also defines a group-aware evaluation protocol. In addition to standard answer accuracy, it uses a score that emphasizes reliable advanced reasoning by rewarding the hardest question in a group only when the prerequisite questions are also correct Xun et al. [2026]. This property makes RTV-Bench a natural testbed for our setting, where we aim to improve not only isolated answer correctness but also consistency across a hierarchy of temporally related questions.

2.3 Grouped Structured Answer Updating

Our method treats answer updating as a structured generation problem. Instead of asking the model to output only a final option letter, we prompt it to answer the full three-question group in a single JSON object. The grouped format makes the hierarchy explicit and encourages the model to resolve all related questions within one temporally grounded response.

For a group with questions q_0 , q_1 , and q_2 , we provide the model with: (i) sampled frames from the video prefix up to the latest timestamp in the group, (ii) the queried timestamp for each question, (iii) the previous queried timestamp within the group, (iv) the question text and answer options, and (v) a short instruction requiring one valid JSON object with one answer and one short rationale per question.

The required output format is:

$$\left\{ \begin{array}{l} \text{"q0": } \{\text{"final_answer": "...", "rationale": "..."}\}, \\ \text{"q1": } \{\text{"final_answer": "...", "rationale": "..."}\}, \\ \text{"q2": } \{\text{"final_answer": "...", "rationale": "..."}\} \end{array} \right. \quad (1)$$

where each `final_answer` is one of the multiple-choice options and each `rationale` is a short grounded sentence. The rationale is not used as a free-form explanation target. Its role is to force the model to associate each predicted answer with concise visual or temporal evidence.

This structured format serves two purposes. At inference time, it makes the model’s behavior easier to inspect than a single option letter. At training time, it defines an intermediate output space that can be constrained and rewarded. This follows the broader intuition that structured intermediate behavior can be improved through reward-based optimization when the reward is aligned with the desired reasoning outcome Shao et al. [2024].

2.4 Video Representation

For each grouped task, we construct the model input from the source video segment spanning time 0 to the latest queried timestamp in that group. We uniformly sample eight frames from this prefix and arrange them into a contact sheet. Each frame is labeled with its approximate timestamp. This representation gives the model a compact view of temporal progression while avoiding the overhead of full video-token processing.

We use the same visual representation for all compared settings. This keeps the comparison focused on the effect of output structure and training rather than on differences in visual input preprocessing. The same grouped prompt content is also preserved across settings except for the required output format.

2.5 GRPO Training

We train the structured model with Group Relative Policy Optimization (GRPO). GRPO is a reinforcement learning method that optimizes generated outputs by comparing multiple sampled completions for the same input and increasing the probability of higher-reward completions Shao et al. [2024]. In our setting, each rollout corresponds to one complete question group.

The reward is designed around two principles. First, malformed outputs should not be rewarded. Second, the optimization target should gradually move from partial question-level correctness to full group-level correctness. We therefore use a hard validity gate followed by a curriculum reward:

$$R_i = \mathbf{1}[\text{JSON valid}]\mathbf{1}[\text{schema valid}] (w_{q0}c_{q0} + w_{q1}c_{q1} + w_{q2}c_{q2} + w_{\text{all}}c_{\text{all}}),$$

$$(w_{q0}, w_{q1}, w_{q2}, w_{\text{all}}) = \begin{cases} (0.15, 0.25, 0.40, 0.20), & p < 0.35 \\ (0.10, 0.20, 0.35, 0.35), & 0.35 \leq p < 0.70 \\ (0, 0, 0, 1.0), & p \geq 0.70 \end{cases}$$

where $c_{qk} \in \{0, 1\}$ indicates whether question qk is answered correctly, $c_{\text{all}} \in \{0, 1\}$ indicates whether all three questions in the group are correct, and p is the normalized training progress. The first two indicator terms enforce that the completion must be valid JSON and must match the exact grouped schema. If either condition fails, the reward is zero.

The curriculum is motivated by the grouped nature of the task. Early in training, dense rewards for individual questions help stabilize learning and prevent the signal from becoming too sparse. As training progresses, the weight shifts toward full group correctness, encouraging the model to solve the three-question hierarchy consistently rather than optimizing each question in isolation. The highest early weight is assigned to $q2$ because it is the most difficult question in the group and the most representative of successful temporal reasoning.

2.6 Training Objective and Model Instantiation

We instantiate the method with Qwen2.5-VL-7B. We train only LoRA adapters on top of the base model while keeping the base weights fixed. The adapters are applied to the attention and MLP projection modules. During training, we sample multiple completions per grouped prompt, score them with the reward above, and update the policy using GRPO.

The resulting method is deliberately simple. It does not introduce a new video encoder or memory module. Instead, it asks whether structured output and reward-based optimization are already sufficient to improve answer updating on a benchmark explicitly built for dynamic video understanding.

3 Experiments

3.1 Experimental Setup

Dataset and grouped split construction. We conduct all experiments on RTV-Bench Xun et al. [2026]. The original benchmark contains multi-timestamp question sets with hierarchical structure, but our method operates on complete three-question groups only. We therefore preprocess each split by grouping questions with the same group identifier and retaining only groups that contain all three levels: $q0$, $q1$, and $q2$. In our setup, each grouped example is the basic unit of both training and evaluation.

Table 1 reports the resulting split statistics. After grouping, the train split contains 602 complete groups, while the dev and test splits each contain 191 complete groups. The number of incomplete groups is small in all three splits, and no malformed group identifiers remain after preprocessing. These grouped examples define the effective dataset used by our method.

Category distributions. We also summarize the semantic and capability-level composition of the benchmark. Table 2 reports counts under the OAE taxonomy (Object, Action, Event), and Table 3 reports counts under the three capability dimensions used in our

Split	Questions	Unique Groups	Complete Groups	Incomplete Groups	$q0$	$q1$	$q2$	Bad QIDs
Train	2823	618	602	16	630	629	1564	0
Dev	874	198	191	7	197	194	483	0
Test	911	195	191	4	200	206	505	0

Table 1: Group-wise split statistics for RTV-Bench. A complete grouped task contains $q0$, $q1$, and $q2$ under the same group ID.

Split	Object	Action	Event
Train	866	1077	880
Dev	282	292	300
Test	251	323	337

Table 2: Question counts by OAE category across train, dev, and test splits.

Split	Perception	Understand	Reason
Train	1036	992	795
Dev	409	255	210
Test	418	324	169

Table 3: Question counts by capability dimension across train, dev, and test splits.

analysis (Perception, Understanding, Reasoning). The OAE categories are relatively balanced across splits. By contrast, the capability dimensions are less balanced: Perception and Understanding are more frequent than Reasoning, especially in the test split. This is relevant when interpreting category-wise gains, since improvements on Reasoning are obtained in the smallest and typically hardest subset.

Model input and grouped evaluation format. For every group, we take the source video segment from time 0 to the latest queried timestamp in that group. We uniformly sample eight frames from this prefix and arrange them into a contact sheet, with each frame labeled by its approximate timestamp. This contact sheet is the visual input to the model in all settings.

The textual input always contains the same three grouped questions, their queried timestamps, their previous queried timestamps within the group, and the multiple-choice answer options. The difference between settings lies in the required output format.

Free-form baseline without JSON. In the free-form setting, the model sees the contact sheet together with the full grouped prompt containing $q0$, $q1$, and $q2$ and their answer choices. It may produce a short natural-language explanation, but its final output must specify exactly one multiple-choice option for each question. Concretely, the model is required to end with a line of the form

Final answers: $q0=A$, $q1=B$, $q2=C$.

Thus, the baseline receives the same grouped visual and textual context as the structured model, but it is not required to produce JSON and it does not provide an explicit rationale field tied to each answer.

Structured JSON setting. In the structured setting, the model sees the same contact sheet and the same grouped multiple-choice questions, but it must return a single JSON object containing one answer and one short rationale for each question:

$$\begin{cases} "q0": \{ "final_answer": "...", "rationale": "...", \\ "q1": \{ "final_answer": "...", "rationale": "...", \\ "q2": \{ "final_answer": "...", "rationale": "... \} \end{cases} \quad (2)$$

The rationale is intended to be a short grounded statement, not a long explanation. This difference is important: compared with the free-form baseline, the JSON model is not only choosing one multiple-choice option per question, but is also forced to organize its predictions question by question and attach a local justification to each one. Our comparison therefore tests whether this added structure improves temporal grounding and grouped consistency.

Compared settings. We evaluate three settings:

1. **Base Qwen2.5-VL-7B, no JSON.** Grouped free-form answering with final multiple-choice predictions.

2. **Base Qwen2.5-VL-7B, JSON.** Grouped structured prompting without additional training.
3. **Trained Qwen2.5-VL-7B, JSON.** Grouped structured prompting after GRPO training with the curriculum reward.

For the trained setting, we report results after both 1 epoch and 5 epochs in order to study the effect of optimization length.

Training details. We instantiate our method with Qwen2.5-VL-7B and train LoRA adapters on top of the frozen base model. LoRA is applied to the attention and MLP projection modules. We use rank 16, LoRA scaling 32, and dropout 0.05. GRPO training uses a learning rate of 1×10^{-5} , weight decay 0, warmup ratio 0.03, per-device batch size 1, gradient accumulation 4, and 4 sampled completions per prompt. The maximum completion length is 512 tokens, with temperature 0.7 and top- p 0.9. We use bfloat16 training when available and enable gradient checkpointing.

Reward design. Our GRPO reward is defined over one grouped rollout. Let c_{q0} , c_{q1} , and c_{q2} be binary indicators for whether the three answers are correct, and let c_{all} indicate whether the entire group is correct. The reward is

$$R_i = \mathbf{1}[\text{JSON valid}]\mathbf{1}[\text{schema valid}] (w_{q0}c_{q0} + w_{q1}c_{q1} + w_{q2}c_{q2} + w_{\text{all}}c_{\text{all}}),$$

$$(w_{q0}, w_{q1}, w_{q2}, w_{\text{all}}) = \begin{cases} (0.15, 0.25, 0.40, 0.20), & p < 0.35 \\ (0.10, 0.20, 0.35, 0.35), & 0.35 \leq p < 0.70 \\ (0, 0, 0, 1.0), & p \geq 0.70 \end{cases}$$

where p is normalized training progress. The first two indicator terms form hard gates: if the completion is not valid JSON or does not match the required schema, the reward is zero. Early in training, the reward gives partial credit for correct answers at the question level, with the largest weight assigned to $q2$ because it is the most difficult question in the group. As training proceeds, the reward shifts toward full-group correctness. In the final phase, the model is rewarded only when all three answers in the group are correct. This curriculum is intended to move the model from stable structured generation to fully consistent grouped reasoning.

Evaluation metrics. We report two main metrics.

Accuracy is the standard per-question multiple-choice accuracy:

$$\text{Accuracy} = \frac{\# \text{ correctly answered questions}}{\# \text{ total questions}}.$$

This metric treats every question independently and measures how often the model predicts the correct option.

Score is a stricter group-level metric:

$$\text{Score} = \frac{\# \text{ groups with } q0, q1, q2 \text{ all correct}}{\# \text{ total groups}}.$$

A group contributes to the score only if the entire triplet is correct. Thus, Score measures grouped consistency rather than isolated answer accuracy. This is the more demanding metric in our setting, because a model must answer all three grouped questions correctly to receive credit.

In addition to overall Accuracy and Score, we report breakdowns by OAE category and by capability dimension. We also analyze the accuracy of the hardest question conditioned on the correctness of the earlier questions in the same group.

Implementation consistency across settings. All three settings use the same grouped data construction, the same contact-sheet representation, and the same grouped prompt content. The only differences are the output format and, for the trained model, the application of GRPO optimization. This controlled setup allows us to attribute performance changes to structured answer updating and reward-based training rather than to changes in the visual input or group construction.

4 Results

Table 4: Evaluation results on RTV-Bench. Perception, Understanding, and Reasoning denote different task categories. Accuracy and score are used as evaluation metrics.

Model	JSON	Training Epochs	Perception Acc (%) / Score	Understanding Acc (%) / Score	Reasoning Acc (%) / Score	Overall Acc (%) / Score
Base Qwen2.5 7B	×	–	34.80 / 7.69	40.10 / 4.69	45.37 / 8.33	38.57 / 6.81
Base Qwen2.5 7B	✓	–	48.72 / 12.09	51.04 / 10.94	43.52 / 2.78	48.52 / 9.95
Trained Qwen2.5 7B	✓	1	52.01 / 17.58	54.17 / 17.19	54.63 / 8.33	53.23 / 15.71
Trained Qwen2.5 7B	✓	5	55.31 / 24.18	56.25 / 15.62	59.26 / 16.67	56.37 / 19.90

We organize the results around three questions. First, does structured grouped JSON prompting improve performance over a comparable free-form baseline? Second, does GRPO training provide gains beyond prompt-only structure? Third, where do the gains appear: in overall accuracy, in full-group consistency, or in the harder semantic and reasoning subsets of the benchmark?

4.1 Does structured prompting help before any training?

Table 4 compares the free-form baseline with the prompt-only JSON setting. The gain is large: overall Accuracy rises from 38.57 to 48.52, and Score rises from 6.81 to 9.95. Since the visual input, grouped questions, and answer choices are identical in both settings, this improvement comes from the structured output requirement rather than from changes in data or input representation.

The gains are strongest in Perception and Understanding. Perception improves from 34.80 to 48.72, and Understanding improves from 40.10 to 51.04. This suggests that requiring one answer and one rationale per question helps the model align its outputs with the grouped task. However, the effect is not uniformly positive. In Reasoning, accuracy drops slightly from 45.37 to 43.52, and Score drops more sharply from 8.33 to 2.78. Structured prompting therefore helps overall, but structure alone is not enough to improve the hardest grouped cases.

4.2 Does GRPO improve performance beyond prompt-only JSON?

Yes. Table 4 shows that GRPO training improves both Accuracy and Score beyond the prompt-only JSON model, and the gains continue from 1 epoch to 5 epochs. After one epoch, overall Accuracy increases from 48.52 to 53.23 and Score from 9.95 to 15.71. After five epochs, the model reaches 56.37 Accuracy and 19.90 Score.

These gains are especially important for Score, which is the stricter metric because it requires the full group to be correct. The increase from 6.81 in the free-form baseline to 19.90 after 5 epochs indicates that the trained model is not only answering more questions correctly, but is also becoming more consistent across grouped question sets. The continued improvement from 1 to 5 epochs also supports the curriculum reward design: the model appears to benefit from the gradual shift from partial question-level correctness to full-group correctness.

4.3 Where do the gains appear across capability dimensions?

Table 4 shows that the gains are broad, but not uniform.

Perception. Perception improves steadily across all settings: 34.80→48.72→52.01→55.31 in Accuracy, and 7.69→12.09→17.58→24.18 in Score. These results indicate that structured answer updating helps even relatively lower-level tasks when those tasks are embedded in a dynamic multi-timestamp setting. The large gain in Score suggests that the trained model does not only perceive the current scene better, but also does so more consistently across grouped questions.

Understanding. Understanding shows the same general trend in Accuracy: 40.10→51.04→54.17→56.25. Score also improves strongly from the baseline to the trained models, although the 5-epoch model is slightly lower than the 1-epoch model on this metric (15.62 vs. 17.19). This is one of the few places where the trend is not strictly monotonic. The most likely interpretation is that longer training still improves answer coverage in this category, but that the trade-off between local correctness and full-group exactness is not fully uniform across dimensions.

Reasoning. Reasoning is where the effect of GRPO is most visible. The prompt-only JSON model underperforms the free-form baseline in this category, especially on Score. But after training, the gains are large: Reasoning Accuracy rises to 54.63 after 1 epoch and 59.26 after 5 epochs, while Reasoning Score increases to 8.33 and then 16.67. This recovery is central to our story. It shows that structure by itself is not enough for the hardest temporal cases, but structure *plus* reward-based optimization is effective. The trained model appears to use the JSON format as a scaffold for grouped reasoning rather than merely as a formatting constraint.

4.4 Does the trained model become more hierarchically consistent?

Table 5: Accuracy Distribution of Question 3 Conditioned on Preceding Question Performance: Q3 Accuracy when at least one of the preceding questions (Q1 or Q2) was answered correctly, versus Q3 accuracy when both preceding questions (Q1 and Q2) were answered incorrectly.

Model	JSON	Training Epochs	Q3 Q1/Q2≥1	Q3 Q1&Q2=0
Base Qwen2.5 7B	×	–	43/127 (33.86)	14/64 (21.88)
Base Qwen2.5 7B	✓	–	54/150 (36.00)	16/41 (39.02)
Trained Qwen2.5 7B	✓	1	66/158 (41.77)	12/33 (36.36)
Trained Qwen2.5 7B	✓	5	85/156 (54.49)	15/35 (42.86)

Table 5 answers this question directly by measuring the accuracy of the hardest question in a group conditioned on the performance of the preceding questions.

For the free-form baseline, Q3 accuracy is 33.86% when at least one earlier question is correct, and 21.88% when both earlier questions are wrong. This already reflects the basic hierarchy of the benchmark: stronger earlier performance is associated with stronger later performance.

The prompt-only JSON model does not strengthen this pattern. Its Q3 accuracy is 36.00% when at least one earlier question is correct, but 39.02% when both earlier questions are wrong. This is the wrong direction. The model is more accurate overall, but the relation between earlier and later questions is not more coherent. This result is important because it shows why prompt-only structure is incomplete: it improves many answers, but does not necessarily align the grouped hierarchy.

GRPO changes this pattern. After one epoch, the expected ordering reappears: 41.77% versus 36.36%. After five epochs, the separation becomes much clearer: 54.49% when at least one earlier question is correct, compared with 42.86% when both earlier questions are wrong. This does not mean that Q3 becomes fully determined by q_0 and q_1 , nor should it. But it does show that the trained model’s hardest answers are more tightly coupled to earlier grouped understanding. This is strong evidence that the model is learning hierarchical consistency rather than only better answer formatting.

4.5 How does the method behave across Object, Action, and Event questions?

Table 6 reports the OAE breakdown. The clearest pattern is that the free-form baseline is weakest on Event questions, with 32.41 Accuracy and 2.78 Score. JSON prompting alone already improves this category substantially, raising Event Accuracy to 47.22 and Event Score to 13.89. GRPO then pushes it further, reaching 51.39 / 19.44 after 1 epoch and 54.63 / 23.61 after 5 epochs.

Table 6: Training evaluation results on the category of OAE: Object, Action and Event. Objects (e.g., players appearing or disappearing during an offensive play), Actions (e.g., dynamic maneuvers by offensive players), and Events (e.g., real-time offensive strategies deployed in the midfield)

Method	JSON	Training Epochs	Object Acc (%) / Score	Action Acc (%) / Score	Event Acc (%) / Score
Base Qwen2.5 7B	×	–	42.48 / 11.76	42.16 / 7.35	32.41 / 2.78
Base Qwen2.5 7B	✓	–	50.33 / 7.84	48.53 / 7.35	47.22 / 13.89
Trained Qwen2.5 7B	✓	1	54.90 / 13.73	53.92 / 13.24	51.39 / 19.44
Trained Qwen2.5 7B	✓	5	58.82 / 21.57	56.37 / 14.71	54.63 / 23.61

This is a meaningful result because Event questions are the highest-level category in the OAE taxonomy. They usually require the model to integrate multiple objects and actions into a scene-level interpretation. The fact that our method delivers the largest score gains in this category suggests that structured answer updating is particularly helpful when the target requires scene-level temporal synthesis.

The gains are not limited to Event questions. Object Accuracy rises from 42.48 to 58.82, and Object Score rises from 11.76 to 21.57. Action Accuracy rises from 42.16 to 56.37, and Action Score rises from 7.35 to 14.71. Thus, the method improves all three OAE categories, but its strongest relative effect appears on the most complex one.

4.6 What overall conclusion do the results support?

Taken together, the three result tables support a consistent conclusion.

First, structured prompting is already useful. When the model is forced to answer grouped questions through a fixed JSON schema with one rationale per question, overall performance improves substantially over a comparable free-form baseline.

Second, prompt-only structure is not enough. It helps Accuracy, but it does not reliably improve the grouped hierarchy, and it can even hurt some harder reasoning metrics.

Third, GRPO resolves this gap. Once the structured output is optimized with a schema-gated curriculum reward, the model improves not only in question-level accuracy but also in full-group correctness, in conditional Q3 consistency, and in complex Event and Reasoning categories.

The most important implication is that better temporal video QA is not only a matter of predicting more correct option letters. It is also a matter of organizing the model’s outputs so that grouped answers are produced in a coherent and temporally grounded way. Our results show that structured answer updating provides a useful interface for this goal, and that reward-based training can turn that interface into a meaningful performance gain.

5 Conclusion

We presented a structured answer-updating framework for dynamic video question answering on RTV-Bench. Our method reformulates grouped multi-timestamp QA as structured generation, requiring one answer and one short rationale for each question in a three-question hierarchy, and then optimizes this behavior with GRPO using a schema-gated curriculum reward. Across all major comparisons, the results support the same conclusion. Structured prompting alone improves overall performance over a comparable free-form baseline, showing that explicit output organization already helps temporal grounding. GRPO training yields further gains, especially on the stricter group-level score, on reasoning-heavy cases, and on event-centric questions. The conditional analysis of the hardest question also shows that training improves hierarchical consistency across grouped questions. These findings suggest that in dynamic video understanding, better performance depends not only on recognizing the current scene, but also on producing answers in a form that can be updated, constrained, and optimized over time.

6 Code

The code for this project can be found [here](#).

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